Algebra and Functions

Questions 1–266

1.01 — Question 1. Without using a calculator, evaluate:

a) $4^{\frac{1}{2}}$
b) $9^{-\frac{1}{2}}$
c) $7^3 \times 7^{-1}$
d) $2 \times 3^3 \times 3^{-2}$
e) $3^5 \div 3^{-1}$
f) $(64^{\frac{1}{3}} \times 16^{\frac{1}{4}})^{-1}$
g) $(\frac{1}{16})^{-\frac{1}{2}}$
h) $(\frac{27}{8})^{-\frac{1}{3}}$
i) $\frac{64^{\frac{2}{3}}}{8^{\frac{5}{3}}}$

1.01 — Question 2. Express the following as a single power:

a) $x\sqrt{x}$
b) $x^2\sqrt{x}$
c) $\frac{x}{\sqrt{x}}$
d) $\frac{x^2}{\sqrt{x}}$
e) $(\sqrt{x})^3$
f) $(\sqrt[3]{x})^2$
g) $\frac{1}{\sqrt{x}}$
h) $\frac{4}{(\sqrt{x})^3}$
i) $\frac{x\sqrt{x}}{\sqrt[3]{x}}$

1.01 — Question 3. Express the following in the form $\sqrt[a]{x^b}$, x^b, $\frac{1}{\sqrt[a]{x^b}}$ or $\frac{1}{x^b}$ as appropriate, where a and b are integers to be found:

a) $x^{\frac{1}{2}}$
b) $x^{-\frac{1}{2}}$
c) x^{-5}
d) $x^{\frac{2}{3}}$
e) $x^{-\frac{3}{4}}$
f) $\frac{1}{x^{-\frac{3}{5}}}$

1.01 — Question 4. Simplify each of the following:

a) $(4a^{10})^{\frac{3}{2}}$
b) $(a^3b^2c)^{\frac{1}{3}} \times (4a^4b^2c)^{\frac{1}{2}}$
c) $(16ab^2)^{\frac{1}{2}} \times (27a^6b^3)^{-\frac{1}{3}}$
d) $(20a^8bc) \div (5a^2b^2c^{-1})$
e) $(3a^2b^3c)^2 \div (9a^2b^{-2}c^4)^{\frac{1}{2}}$

1.01 — Question 5. Simplify each of the following:

a) $\frac{(2a)^3\times(3a^2)^5}{(6a^5)^2}$

b) $\frac{(3a)^3\times(4a^2)^2}{(2a^4)^3}$

c) $\left(\frac{a^{\frac{1}{2}}\times a^{\frac{5}{2}}}{(a^3)^4}\right)^{-1}$

d) $(\frac{32}{a^8})^{\frac{3}{5}} \times (\frac{a^7}{4})$

1.01 — Question 6. Simplify each of the following:

a) $\sqrt{x^3 \times x^5}$

b) $\sqrt{x^{\frac{3}{5}} \times x^{\frac{1}{5}}}$

c) $\sqrt{49x^6y^4} \times (27x^3y^9)^{-\frac{1}{3}}$

d) $\sqrt{16x^4y^6} \times (9x^2y^3)^{-\frac{1}{2}}$

1.01 — Question 7. Give each of the following as sums of terms in the form ax^n:

a) $\sqrt{x}(2x+1)$
b) $(\sqrt{x}+1)(\sqrt{x}-2)$
c) $\frac{x+1}{\sqrt{x}}$
d) $\frac{x^2-1}{\sqrt{x}}$
e) $\frac{4}{x}+\frac{1}{\sqrt{9x}}$
f) $\frac{\sqrt{x}-3}{x\sqrt{x}}$

1.01 — Question 8. A right-angled triangle has perpendicular sides $7a^{\frac{1}{2}}$ and $24a^{\frac{1}{2}}$. Find

a) the length of the hypotenuse side.
b) the area of the triangle.

1.01 — Question 9. Solve each of the following:

a) $x^{\frac{1}{2}} = 2$
b) $x^{-\frac{1}{2}} = 3$
c) $x^{\frac{1}{3}} = \frac{1}{2}$
d) $x^{-\frac{1}{2}} = 11$
e) $x^{-\frac{3}{2}} = \frac{8}{27}$
f) $2x^{-\frac{1}{3}} = 3$
g) $x^{-\frac{2}{3}} = 2\frac{7}{9}$

1.01 — Question 10. Mo was asked to find the solutions to

$$x^{\frac{7}{4}} - 2187 = 0.$$

His solution is shown below.

$$\begin{aligned} x^{\frac{7}{4}} &= 2187 \\ x^{\frac{1}{4}} &= \sqrt[7]{2187} = \pm 3 \\ x &= 3^4 \text{ or } (-3)^4 \\ x &= 81 \text{ or } -81 \end{aligned}$$

Find his mistake(s) and write a correct solution.

1.01 — Question 11. Write each of these in the form 8^k:

a) $\sqrt{8}$
b) $8^{10} \times 8^{-12}$
c) $\frac{1}{64}$
d) $\sqrt[3]{64}$
e) $\frac{2}{\sqrt{8}}$
f) $\frac{2^6}{\sqrt{8}}$
g) $\sqrt[3]{16} \times (8^3)^{\frac{1}{9}}$
h) $2 \times \sqrt{64} \times \sqrt[3]{16}$

1.01 — Question 12. Solve each of the following equations:

a) $2^x \times 2^{\frac{1}{2}} = 2$
b) $5^x \times \sqrt{5} = \frac{1}{25}$
c) $3^{x+1} = 9^x$
d) $4^{2x} = \sqrt{2}$
e) $2^{3x-1} = 32$
f) $2^{3x+2} = 4^{x+5}$
g) $9^{2x-1} = \frac{1}{\sqrt{3}}$
h) $\frac{9^x}{3^{x+2}} = 27$
i) $3^{x^2-x} = 729$

1.01 — Question 13. Simplify each of the following into a single term (selecting the smaller base):

a) $4^x + 4^x + 4^x + 4^x$

b) $2^{2x} + 4^x$

c) $3 \times 2^{3x} + 5 \times 8^x$

d) $5 \times 16^{2x} - 3 \times 2^{8x}$

e) $3^{2x+1} + 3^{2x}$

f) $8 \times 2^x + 3 \times 2^{x+1}$

g) $2^{2x+4} + 4^x$

1.01 — Question 14. Simplify each of the following so that there are only positive exponents in the numerator and denominator:

a) $\frac{x+x^{-2}}{x^2y+x^{-1}}$

b) $\frac{x^2y^{-1}+x^{-1}}{xy^2+x^{-2}}$

c) $\frac{x^4-x^{-2}yz^3}{x^2+z^{-1}\sqrt{y}}$

d) $\frac{x^5-x^{-1}yz^4}{x^{-3}+z^2y^{-\frac{1}{2}}}$

1.01 — Question 15. Solve each of the following equations:

a) $4^{2x} \times 3^{2x} = 12$

b) $5^{x} \times 2^{2x} = 20$

c) $3^{2x} \times 9^{x} = 9$

d) $\frac{2^{x}}{2^{\frac{x}{2}} \times 2^{4}} = 1$

1.01 — Question 16. Find the solutions to $4^{x+2} - 2^{2x+3} - 16 = 0$.

1.01 — Question 17. Find the value of k such that $\frac{2^{501}-2^{497}}{30} = 64^k$.

1.01 — Question 18. Solve $\frac{2^x}{32\times 2^{\sqrt{x}}} = 2$.

1.01 — Question 19. Find one solution to $(x^2 - 4x - 1)^{3x+2} = \frac{1}{4}$.

1.02 — Question 1. We would expect completion of these questions without the use of a calculator. Simplify

a) $\sqrt{32}$
b) $2\sqrt{12}$
c) $\sqrt{45}$
d) $\sqrt{200}$
e) $-4\sqrt{50}$

1.02 — Question 2. Simplify

a) $\sqrt{5} \times \sqrt{2}$
b) $\sqrt{5}\sqrt{15}$
c) $\sqrt{8} \times 3\sqrt{2}$
d) $\sqrt{\frac{1}{9}}$
e) $\frac{\sqrt{28}}{\sqrt{14}}$
f) $\sqrt{25x^2}$

1.02 — Question 3. Simplify

a) $2\sqrt{5} - \sqrt{5}$
b) $\sqrt{8} + 2\sqrt{2}$
c) $\sqrt{12} - \sqrt{3}$
d) $\sqrt{128} - \sqrt{72}$
e) $3\sqrt{12} + \sqrt{27}$
f) $\sqrt{150} - \sqrt{96}$
g) $\sqrt{3} + \sqrt{27} + \sqrt{12}$
h) $\sqrt{350} + \sqrt{24}$
i) $\sqrt{32a} + 2\sqrt{a} - \sqrt{28a}$

1.02 — Question 4. Simplify

a) $\sqrt{2} \times \sqrt{7} \times \sqrt{3}$
b) $\sqrt{18} \times \sqrt{2}$
c) $2\sqrt{3} \times 2\sqrt{3}$
d) $5\sqrt{3} \times 2\sqrt{5}$
e) $-3\sqrt{3} \times 2\sqrt{3}$
f) $(2\sqrt{6})^2$
g) $(3\sqrt{8})^2$
h) $(2\sqrt{5})^3$

1.02 — Question 5. Express each of the following in the form $\sqrt{a}$ where a is a positive integer:

a) $2\sqrt{5}$
b) $4\sqrt{3}$
c) $3\sqrt{2}$
d) $2\sqrt{7}$
e) $2\sqrt{43}$

1.02 — Question 6. Express the following as a single square root:

a) $a\sqrt{b}$
b) $2\sqrt{a}$
c) $(\sqrt{a})^5$
d) $(a\sqrt{b})^3$

1.02 — Question 7. Express $\sqrt{525}$ in the form $p\sqrt{q}\sqrt{r}$ where p, q and r are prime numbers.

1.02 — Question 8. Write this ratio in its simplest form: $\sqrt{300} : \sqrt{48} : \sqrt{27}$.

1.02 — Question 9. Rationalise the following surds:

a) $\frac{3}{\sqrt{2}}$
b) $\frac{6}{\sqrt{3}}$
c) $\frac{6}{\sqrt{10}}$
d) $\frac{2}{\sqrt{6}}$
e) $\frac{10}{\sqrt{10}}$
f) $\frac{4\sqrt{2}}{\sqrt{8}}$

1.02 — Question 10. Expand and simplify the following:

a) $\sqrt{5}(1+\sqrt{5})$
b) $\sqrt{11}(2+\sqrt{11})$
c) $4(\sqrt{3}+2\sqrt{5})$
d) $(\sqrt{3}+1)(\sqrt{3}-1)$
e) $(\sqrt{6}+1)(2+\sqrt{6})$
f) $(\sqrt{8}-1)(2+\sqrt{8})$
g) $(\sqrt{24}+1)(\sqrt{2}-3)$

1.02 — Question 11. Rationalise the following surds:

a) $\frac{1}{\sqrt{2}+1}$
b) $\frac{3}{\sqrt{5}-2}$
c) $\frac{2}{1-\sqrt{6}}$
d) $\frac{1}{a+\sqrt{b}}$
e) $\frac{1}{a-\sqrt{2b}}$

1.02 — Question 12. Simplify the following, giving your answers in the form $p\sqrt{q}$ where p and q are non-zero rational numbers:

a) $\sqrt{27} + \frac{15}{\sqrt{3}}$

b) $\frac{\sqrt{54}}{2} - \frac{10}{\sqrt{6}}$

c) $\frac{\sqrt{150}}{5} + \frac{30}{\sqrt{24}}$

1.02 — Question 13. Solve $x(4 - \sqrt{6}) = 10$ giving your answer in the form $a + b\sqrt{6}$, where a and b are non-zero integers to be found.

1.02 — Question 14. Rationalise the denominator and simplify the following:

a) $\frac{\sqrt{3}}{\sqrt{2}-2}$

b) $\frac{2\sqrt{5}-1}{\sqrt{5}+4}$

c) $\frac{5\sqrt{3}-2}{2-\sqrt{7}}$

d) $\frac{2\sqrt{3}-7}{2\sqrt{3}-3}$

e) $\frac{1-\sqrt{k}}{2+\sqrt{2k}}$

1.02 — Question 15. Solve $x(1-\sqrt{2})+\sqrt{2}=5$ giving your answer in the form $a+b\sqrt{2}$, where a and b are non-zero integers to be found.

1.02 — Question 16. Show that $\frac{3\sqrt{10}}{\sqrt{2}+\sqrt{5}}$ may be written in the form $p\sqrt{q}-q\sqrt{p}$ where p and q are non-zero integers to be found.

1.02 — Question 17. Show that $\frac{\sqrt{3}+\sqrt{7}}{\sqrt{3}-\sqrt{7}}$ may be written in the form $a+b\sqrt{21}$ where a and b are non-zero constants to be found.

1.02 — Question 18. The expression $\frac{a+\sqrt{6}}{1-\sqrt{6}}$ may be written in the form $p + q\sqrt{6}$. Find the values of p and q in terms of a.

1.02 — Question 19. Fully simplify the expression $\frac{12}{\frac{1}{\sqrt{3}}+\sqrt{3}}$.

1.02 — Question 20. A cylinder has a height of $(\sqrt{3}+1)$cm and a radius of $(\frac{1}{\sqrt{3}-2})$cm. Show that the volume of the cylinder is exactly $\pi(19+11\sqrt{3})\text{cm}^3$.

1.02 — Question 21. The expression $\frac{-k+2\sqrt{3}}{\sqrt{27}+1}$ may be written in the form $a+b\sqrt{3}$. Find the values of a and b in terms of k.

1.02 — Question 22. The expression $\frac{2-\sqrt{k}}{1+\sqrt{k}}$ may be written in the form $a+b\sqrt{k}$. Find the values of a and b in terms of k.

1.02 — Question 23.

a) Simplify $\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}}$.

b) Hence write down the solution to $\frac{\sqrt{7}-\sqrt{6}}{\sqrt{7}+\sqrt{6}}$.

1.02 — Question 24. Evaluate the sum $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \cdots + \frac{1}{\sqrt{99}+\sqrt{100}}$.

1.02 — Question 25. By using algebraic methods, show that $\sqrt{2} + \sqrt{3} > \sqrt{5}$.

1.02 — Question 26. Show that $\frac{2}{\sqrt{5}+\sqrt{2}+1} = 3 + 2\sqrt{2} - \sqrt{5} - \sqrt{10}$.

1.05 — Question 1. For each of the following situations, write an equation to model the information:

a) When making pastry, the weight of fats (y) needed is half the weight of the flour (x)

b) A business adds a tax of 20% to all items it sells (x) before calculating its sales price (y)

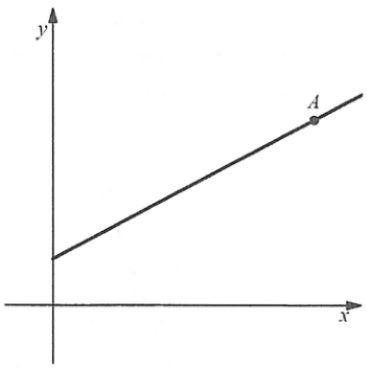
c) For each £1 converted into US dollars (x) you receive \$1.29 (y)

1.05 — Question 2. Three phone companies charge mobile phone tariffs usually known as 'pay-as-you-go' which charge £C the user per t minutes and have no monthly cost.

Phone Company	Standard Charge (per minute)
H_2O	25p
UU	30p
PiffPaff	24p

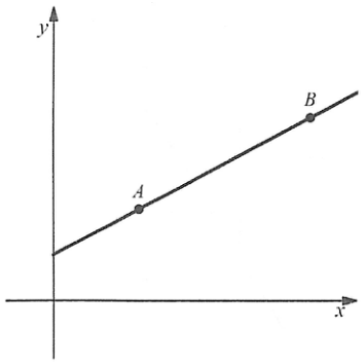
a) On the same axes, sketch a graph representing the cost over time for making calls using each of the companies. Label the point at which £3 has been spent.
b) In each case write down an equation for C in terms of t.

1.05 — Question 3. The graph below shows a straight-line graph which goes through the point A.



Find the equation of the line given
a) Point A is $(1, 9)$ and the y-intercept is 8
b) Point A is $(5, 4)$ and the y-intercept is 2
c) Point A is $(3, 7)$ and the y-intercept is 5

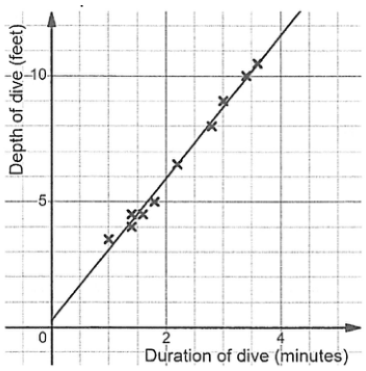
1.05 — Question 4. The graph below shows a straight-line graph going between two points A and B.



Find the equation of the line in the form $y = mx + c$ given points A and B are

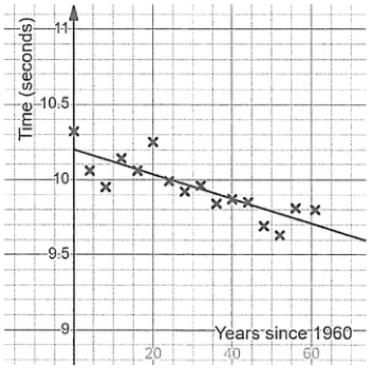
a) $(2, 10)$ and $(5, 19)$
b) $(1, 2.5)$ and $(7, 5.5)$
c) $(3, 7)$ and $(15, 12)$

1.05 — Question 5. Data has been collected on the dive of penguins. The depth of their dives d in feet and duration of their dive duration in time t minutes is plotted below.



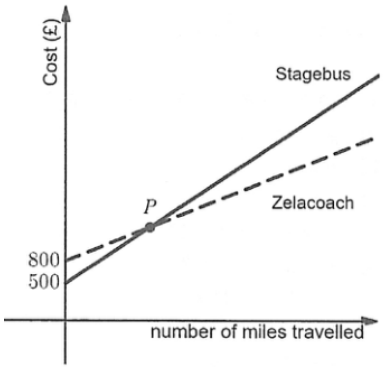
a) Use the graph to write an equation of d in term of t
b) Use your equation to predict how deep a penguin will dive in a 2.5 minute dive.
c) What does the gradient of the graph represent? State the units.
d) What does the vertical intercept represent in context?
e) Comment on the validity of the vertical intercept.

1.05 — Question 6. The graph below shows the times of the gold medallists, T seconds, in the men's 100 metre event in the Olympics against the years since 1960, x years. The line is given by the equation $T = -0.0082x + 10.2$.



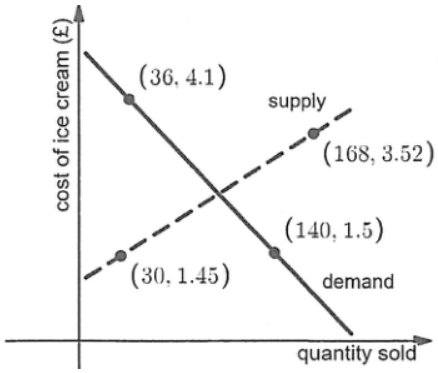
a) According to the model, what time will someone in the 2028 Olympics achieve?
b) Explain what the 10.2 and the -0.0082 represents in context.
c) Explain why a linear model is not valid in the long term.

1.05 — Question 7. The graph below shows the cost £C of hiring a coach for the number of miles travelled m from two different companies, Stagebus and Zelacoach.



For a distance of 220 miles, Stagebus quotes a cost of £2000 whilst Zelacoach charges £1700.
a) Explain what the values on the vertical axis represent in context for each company.
b) Find an equation that models the costs of hiring coaches from the two companies.
c) Identify the coordinates of the point P labelled on the graph and state the relevance of this point in context.

1.05 — Question 8. The graph below shows the link between the price of an ice cream £P and the number of them sold x in the shop.



State the point at which the two lines intersect and interpret what that point means in context.

1.05 — Question 9. Two companies charge different costs for electrical work at their business premises. **Electrosolutions:** A call out charge plus £30 an hour after that. **Ohms Electrical:** Starting at £90 covering the first 2 hours and then following a linear relationship with an hourly rate. Rebecca uses both companies for her business and finds that they both charged £170 for 4 hours work.

a) Find the call out charge of Electrosolutions.
b) Find the hourly rate of Ohms Electrical.
c) Ohms Electrical want to change to a call out charge with their current hourly rate. What would be their initial call out charge?

1.05 — Question 10. A bakery makes muffins. They sell the muffins at £2.00 each. There is an upfront cost in making the muffins, plus a fixed cost per muffin. When 500 muffins are made and sold, a profit of £300 is made, but when 100 muffins are made and sold, there is a loss of £50.

a) Show that the cost £C in terms of the number of muffins x, may be modelled as $C = 1.125x + 137.5$.

b) Find the minimum number of muffins that need to be made and sold for the bakery to not make a loss.

1.06 — Question 1. Algebraic Manipulation and Solving Equations Fully factorise the following:

a) $x^2 + 14x + 45$

b) $x^2 + 3x - 18$

c) $x^2 - 7x$

d) $2x^2 + 11x - 6$

e) $2x^2 - 17x + 35$

f) $8x^2 - 14x + 3$

g) $6x^2 + 17x + 5$

h) $-2x^2 - x + 3$

i) $x^2 - 9$

j) $4x^2 - 5$

1.06 — Question 2. Fully factorise the following where a and b are constants:

a) $x^2 - a^2$

b) $(ax)^2 - b^2$

c) $x^2 - a$

1.06 — Question 3. Fully factorise the following:

a) $8(x+2) - x(x+2)$

b) $-(x-2) + 2x(x-2)$

c) $12(x-5)^2 + (x-5)$

d) $(x-7)^2 - 4$

e) $\frac{2}{3}(3x-10)^2 - \frac{32}{3}$

1.06 — Question 4. Solve the following:

a) $x^2 = x$
b) $x^2 - 4 = 0$
c) $x^2 - 8 = 0$
d) $x^2 + 2x = 0$
e) $(x-5)^2 = 0$
f) $x^2 - 12x + 36 = 0$
g) $\frac{1}{5}x^2 - \frac{1}{5}x = 4$
h) $(x+3)^2 = 6$
i) $(x-1)(x-3) = 8 - 2x$

1.06 — Question 5. Solve the following, giving your answers in exact form:

a) $x^2 - 8x + 5 = 0$
b) $2x^2 - 40 = 0$
c) $2x^2 + 16x + 24 = 0$
d) $3x^2 - 5x - 1 = 0$
e) $ax^2 - b = 0,\ a \neq 0$
f) $ax^2 - bx = 0,\ a \neq 0$

1.06 — Question 6. Two students are trying to find the solutions to $(x-5)^2 = (x-5)$. In each case, explain the mistake they made and then show your workings for correctly solving it.
Student 1:
When $x = 5$, $(5-5)^2 = (5-5)$
so $0 = 0$ which means $x = 5$
Student 2:
Divide both sides by $x - 5$: $x - 5 = 1$
so $x = 6$

1.06 — Question 7. Solve $x - 3 = 2\sqrt{x}$.

1.06 — Question 8. Solve $2x + 2 = 5\sqrt{x}$.

1.06 — Question 9. By using the substitution $p = x^2$ find the solutions to $2x^4 - 11x^2 + 5 = 0$.

1.06 — Question 10. By using the substitution $p = x^3$ find the solutions to $3x^3 + 19 = \frac{40}{x^3}$.

1.06 — Question 11. By using the substitution $p = 2x^2 - x$, find the solutions to $12x^4 - 12x^3 + 5x^2 - x - 2 = 0$.

Completing the Square

1.06 — Question 12. Complete the square on each of the following:

a) $y = x^2 - 8x$
b) $y = x^2 + 6x$
c) $y = x^2 - 10x + 40$
d) $y = x^2 + 8x - 20$
e) $y = x^2 + 5x + 7$
f) $y = 2x^2 - 16x - 20$
g) $y = 4x^2 - 24x + 25$
h) $y = 3x^2 - 7x + 1$

1.06 — Question 13. Complete the square on each of the following:

a) $y = -x(x - 2)$
b) $y = (x + 3)(x + 2)$
c) $y = -2x^2 - 4x + 5$
d) $y = -2x^2 - 7x + 1$
e) $y = -3x^2 + 5x + 10$
f) $y = 25 - 4x^2$
g) $y = x^2 + px + r$

1.06 — Question 14. Explain the graphical relevance of a and b in the curve $y = (x - a)^2 + b$.

1.06 — Question 15. For each of the following curves, find the coordinates of the vertex and state the type of vertex:

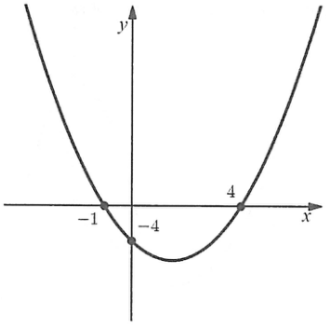
a) $y = (x - 2)^2 - 12$
b) $y = (x + 2)^2 - 8$
c) $y = -(x - 1)^2 + 12$
d) $y = -(x + 2)^2$
e) $y = 3(x + 4)^2 + 5$
f) $y = 2(x + \frac{5}{2})^2 + \frac{1}{2}$
g) $y = -10(x - 4)^2 - 100$
h) $y = x^2 - 6x - 2$
i) $y = 3x^2 + 6x + 5$
j) $y = -2x^2 - 4x + 3$
k) $y = 2x^2 - 5$
l) $y = -5x^2 + 12$
m) $y = a(x - b)^2,\ a < 0$
n) $y = x^2 + px + r$

1.06 — Question 16. State the line of symmetry in the following curves:

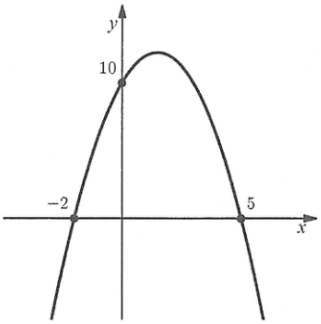
a) $y = (x-4)^2 + 5$
b) $y = 2(x+3)^2 + 7$
c) $y = -3(x-1)^2 - 6$
d) $y = (x-2)(x+4)$
e) $y = 2(x-1)(x-6)$
f) $y = 5(x+2)(x-8)$
g) $y = (x-a)(x-b)$
h) $y = (x-p)^2 + q$

1.06 — Question 17. In each case, find the equation of the curve in the form $y = (x-a)(x-b)$ and state the coordinates of the vertex:

a)

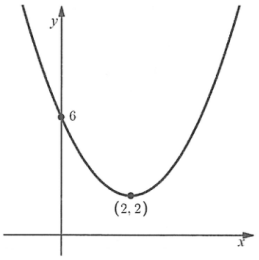


b)

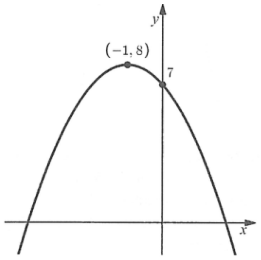


1.06 — Question 18. For each of the following functions, find the equation of the graph in the form $y = a(x-b)^2+c$ where a, b and c are values to be found:

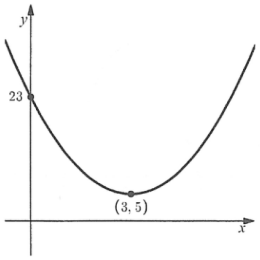
a)



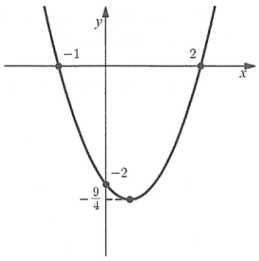
b)



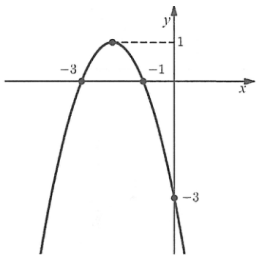
c)



d)



e)



1.06 — Question 19. Find the equation of the curve in the form $y = ax^2 + bx + c$ for the curves described below:

a) Minimum point at $(1, 5)$, intersecting the y-axis at $y = 6$
b) Minimum point at $(3, -1)$, intersecting the y-axis at $y = 17$
c) Maximum point at $(-2, 4)$ intersecting the y-axis at $y = -76$

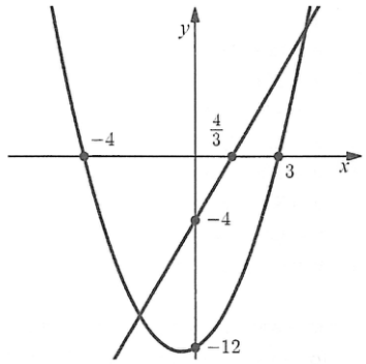
1.06 — Question 20. Simultaneous Equations
We suggest that Algebra: Simultaneous Equations is helpful with Q20-Q23 Find the solutions to following pairs of simultaneous equations:

a) $y = 2x + 3$, $y = x^2 + x + 1$
b) $y = -2x + 3$, $y = x^2 - 2x + 2$
c) $y = 2 - x$, $y = 6 - x^2 - x$
d) $y = -2x^2 + x + 4$, $y = -x^2 + x + 3$
e) $y = x^2 - 2x$, $y = 2x^2 + x - 4$

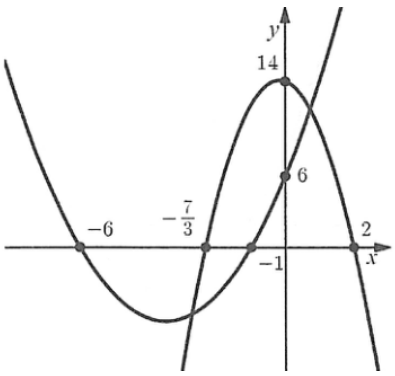
1.06 — Question 21. Find the point(s) of intersection of the following curves and straight lines:

a) $y = x^2 + 1$, $y = \frac{1}{2}x + 1$
b) $y = -x^2 + 5$, $y = -x + 3$
c) $y = x^2 - 4x + 5$, $y = 2x - 4$
d) $y = 2x^2 - 4x + 3$, $y = 4x - 5$

1.06 — Question 22. Find the points of intersection of the quadratic curve and the straight line below:



1.06 — Question 23. Find the points of intersection of the quadratic curves shown in the graph below:



1.07 — Question 1. Solve the following sets of simultaneous equations:

a) $2x + 3y = 5$ and $7x - y = 6$

b) $6x = 5 - 2y$ and $x - 2y = 2$

c) $\frac{x-9}{3} + y = 0$ and $2(x + y) = 2$

d) $3x - y = 4$ and $\frac{y}{2} + \frac{x}{4} = 5$

1.07 — Question 2. It is given that $y - kx = 10$ and $y = 2x - 10$, where k is an unknown constant, intersect when $y = 15$. Find

a) the x-coordinate of the point of intersection,

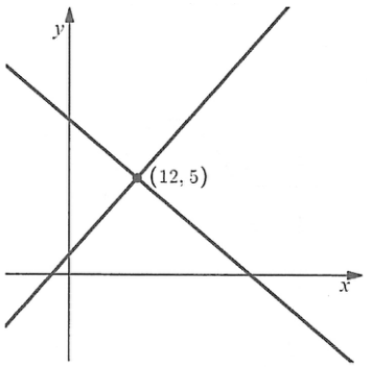
b) the value of k

1.07 — Question 3. Two lines are given to be $y + ax = 5$ and $\frac{y}{b} + \frac{11x}{80} = 1$ where a and b are unknown constants. The two lines never intersect, and one line passes through the y-axis at $y = 8$.

a) the value of b

b) the value of a

1.07 — Question 4. The diagram below shows two lines, which intersect at the point $(12, 5)$.



Given one line has gradient $-\frac{1}{4}$ and the other line intersects the y-axis at $y = 1$, find the equations of both lines in the form $ay + bx = c$ where a, b and c are integers.

1.07 — Question 5. Solve the following sets of simultaneous equations:

a) $y = 2x + 4$ and $y = x^2 + 1$

b) $y = 3x - 2$ and $y = x^2 - 4x - 10$

c) $y = \frac{1}{2}x$ and $y = x^2 + 2xy$

1.07 — Question 6. Two equations are given by $3y - x + 10 = 0$ and $x^2 + y^2 = 10$.

a) Show that the two equations may be satisfied by $x^2 - 2x + 1 = 0$.

b) Hence solve the simultaneous equations $3y - x + 10 = 0$ and $x^2 + y^2 = 10$.

1.07 — Question 7. Solve the following sets of simultaneous equations:

a) $y + 3x = 15$ and $y^2 = 9(x - 3)$

b) $y - x = 4$ and $x^2 + y = 10$

c) $y - 2x = 12$ and $x^2 + 2xy = 5$

1.07 — Question 8. Solve the following sets of simultaneous equations:

a) $y + x = 4$ and $x^2 + y^2 = 16$

b) $y + 4x = 6$ and $x^2 + 2x + y^2 - 3y = 1$

c) $2y + x = 8$ and $x^2 + y^2 - xy = 12$

d) $x = 2y + 1$ and $y^2 + 8 = 3xy$

1.07 — Question 9. Two equations are given by $x^2 + 2x + y^2 = 24$ and $2y = x - 12$. Show that there are no solutions when solving simultaneously.

1.07 — Question 10. The curve $y = px^2 - qx + 1$ where p and q are positive constants, passes through the points $(5, 36)$ and $(-2, 15)$. Find the values of p and q.

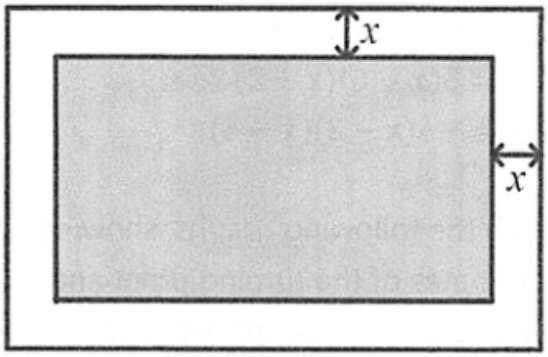
1.07 — Question 11. The curve $y = px^3 - 5x^2 + x + q$ where p and q are positive constants, passes through the points $(1, 6)$ and $(-\frac{1}{2}, \frac{21}{4})$. Find the values of p and q.

1.07 — Question 12. Find the two positive real numbers with a product of 20 and a difference of $\frac{8}{3}$.

1.07 — Question 13. Solve the simultaneous equations $2^{x-1} = 4^{2y}$ and $9^{x-2} = 3^{y+1}$.

1.07 — Question 14. Solve the simultaneous equations $5^{2x-3} = 25^{y-1}$ and $16^{x-1} = 2^{3y}$.

1.09 — Question 1. A rectangular garden is 30m by 40m. The path surrounding a rectangular pond has a consistent width of x and total area of 864m^2. Find the width and length of the pond.



1.09 — Question 2. A room is comprised of two areas; one square with length x m, and one rectangular with width $(x + 2)$ m and length 10 m. It has a total area of 44 m^2. Find the value of x.

1.09 — Question 3. For the equations below, without manipulating the equation, state the number of real roots:

a) $y = (x+1)^2 + 2$
b) $y = -(x+1)^2 + 2$
c) $y = (x-4)^2 - 3$
d) $y = -(x-4)^2 - 3$
e) $y = (x+2)^2$

1.09 — Question 4.

a) Sketch the graph of $y = \frac{3}{4}x(x-1)$ for $0 \leq x \leq 4$.
b) Find the minimum of the curve.

1.09 — Question 5.

a) Sketch the graph of $y = \frac{22}{25}x(200 - x)$ for $0 \leq x \leq 200$.

b) Find the maximum of the curve.

1.09 — Question 6. Sketch the graph of $y = a(x - b)^2 + c$ for constants a, b and c in each of the cases below. Label where the curve intersects with the axes.

a) $a > 0, b > 0, c > 0$

b) $a > 0, b < 0, c > 0$

c) $a < 0, b > 0, c < 0$

d) $a < 0, b < 0, c < 0$

1.09 — Question 7. For the graph of $y = a(x-b)^2 + c$ where a, b and c are constants, state the conditions for the graph to have a minimum in the top left quadrant and the y-intercept to be positive.

1.09 — Question 8. For the graph of $y = a(x-b)^2 + c$ where a, b and c are constants, state the conditions for the graph to have a maximum in the top right quadrant and the y-intercept to be negative.

1.09 — Question 9. The graph of $y = x^2 + 3x + k$ where k is a constant has a minimum with coordinates $(-\frac{3}{2}, -\frac{1}{2})$. Find the coordinates of where the curve intersects

a) the y-axis.

b) the x-axis.

1.09 — Question 10. The graph of $y = -x^2 + 4kx + k$ where k is a constant has a maximum with coordinates $(2, 5)$. Find the coordinates of the points where the curve intersects

c) the y-axis.

d) the x-axis.

1.09 — Question 11. By completing the square, show that the solutions to $ax^2 + bx + c = 0$ are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

1.09 — Question 12. A curve has equation $y = 2x^2 + x + k$ where $k \neq 0$. Given that the roots of the equation have a difference of 2.5, find the value of k.

1.09 — Question 13. A curve has equation $y = -5x^2 - kx + 3$ where $k \neq 0$. Given that the roots of the equation have a difference of 1.6, find the possible values of k.

1.10 — Question 1. Calculate the discriminant:

a) $x^2 - 2x - 6$
b) $x^2 - 9$
c) $-x^2 + x + 5$
d) $-0.5x^2 + 12$
e) $2x^2$
f) $2x^2 - 3x + 2$
g) $-3x^2 + 3x - 1$
h) $x^2 - kx + 2$

1.10 — Question 2. State the number of roots for each of the following functions:

a) $f(x) = x^2 - 4x + 5$
b) $f(x) = x^2 + x - 6$
c) $f(x) = 2x^2 - 2x - 4$
d) $f(x) = x^2 - 3x + 8$
e) $f(x) = x^2 + 8x + 16$
f) $f(x) = -2x^2 + 5x + 3$
g) $f(x) = -4x^2 + 12x - 9$

1.10 — Question 3. Show that $3x = 3x^2 + 1$ has no real solutions.

1.10 — Question 4. Find the values of k for each of the following equations with equal roots:

a) $4x^2 + kx + 9 = 0$
b) $3x^2 + kx + 12 = 0$
c) $x^2 - 3x + 2 + \frac{1}{k} = 0$
d) $x^2 + (2k + 10)x + k^2 + 5 = 0$
e) $16x^2 + 4x + k^2 = 0$

1.10 — Question 5. The equation $x^2 + 3px + p = 0$ where p is a non-zero, has equal roots. Find the value of p.

1.10 — Question 6. Show that the discriminant of $(x-p)(x-q)$ is $(p-q)^2$.

1.10 — Question 7. Show that the discriminant of $k(x-p)^2+q$ is $-4kq$.

1.10 — Question 8. The curve $y = -\frac{1}{2}x^2 + kx + k$ touches the line $2y - 2x = 5$ at one point. Find the possible value(s) of k.

1.10 — Question 9. The line $5y + x = k$ touches the curve $2x^2 + 2y = 1$ once.

a) Find the value of k.

b) Hence find the values at which $5y + x = k$ intersects the x-axis and the y-axis.

1.10 — Question 10. A function $f(x) = ax^2 + bx + c$ has constants a, b and c such that $a < 0$, $b > 0$ and $c < 0$. The discriminant of $f(x)$ is 0. Sketch the graph of $f(x)$. Label the intersections with the axes.

1.10 — Question 11. Show that $y = x^2 + kx - x$ for $k > 0$, always has two real roots.

1.10 — Question 12. For the equations $2x + y = 1$ and $x^2 - 4ky + 5k = 0$ where k is non-zero constant,

a) Show that $x^2 + 8kx + k = 0$.

b) Given that $x^2 + 8kx + k = 0$ has equal roots, find the value of k.

c) Hence for this value of k find the solution of the simultaneous equations.

1.10 — Question 13. Find the relationship between non-zero constants p and q, such that the equation $y = px^2 + (p + q)x + q$ has equal roots.

1.10 — Question 14. The curves $f(x) = x^2 - kx - 4$ and $g(x) = -x^2 + x + 4k$ intersect once.

a) Find the possible value(s) of k.

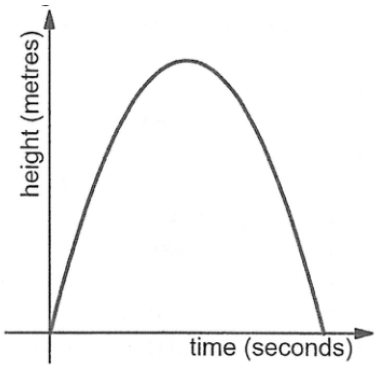
b) Hence find the coordinates of the point(s) of intersection of the two curves.

1.10 — Question 15. The equation $x^4 - 6x^2 + k = 0$ has two pairs of equal roots. Find the value of k.

1.11 — Question 1. A ball is thrown vertically upwards with a speed of 19.6ms^{-1}. The height of the ball, h metres, may be modelled by the equation $h = 19.6t - 4.9t^2$ where t is the time given in seconds.

a) Find the height of the ball after 2 seconds.
b) Find the times at which the ball is at a height of 14.7 metres above the ground.
c) Explain what your two solutions in the previous part represent.

1.11 — Question 2. A ball is hit into the air, the graph below shows its vertical height above the ground before it hits the ground.



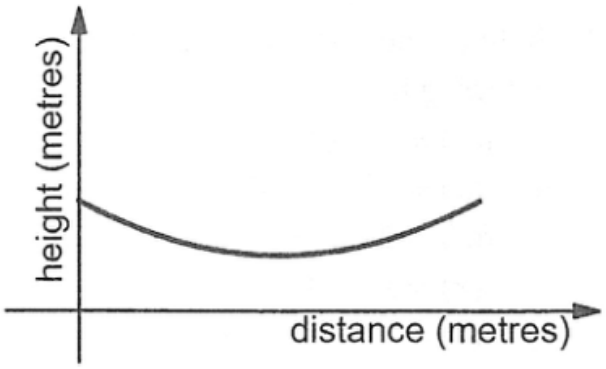
A model for the height of the ball, h metres at time t seconds is given by $h(t) = 10t - 2t^2$. Find

a) the time when the ball hits the ground.
b) $h(t)$ in the form $k(t-p)^2 + q$ where p, q and k are constants to be found.
c) the maximum height the ball reaches and the time at which it reaches the maximum height.

An improved model starts the ball at $h = 1$.

d) State the new model in the form $h(t) = at^2 + bt + c$.
e) State the coordinate of the maximum height of the curve of this new model.

1.11 — Question 3. Two identical fence posts are 2.5 metres apart and a rope is hung between them. The curve $y = \frac{1}{100}x^2 - \frac{1}{40}x + \frac{1}{2}$ may be used to model the rope, where y is the vertical distance in metres above the ground and x is the horizontal distance in metres from the left post. The curve is shown in the diagram below:



a) State the height of the posts.
b) Rearrange the model into the form $a(x-b)^2 + c$ where a, b and c are constants to be found.
c) State the horizontal and vertical position of the minimum height of the rope.
d) State why the model is not effective for $x > 2.5$.

1.11 — Question 4. The path of a swimmer may be modelled by $d = 0.1x^2 - x + 0.9$ for $x \geq 0$ where d is the depth of the swimmer in relation to the water surface and x metres is the distance from the poolside.

a) Sketch the graph of the model.

According to the model:

b) Find the depth that the swimmer swims below the surface of the water.
c) Determine the horizontal distance the swimmer swam under the water.
d) State how far from the pool edge the swimmer is when they first contact the water.
e) From what value of x does the function become an unrealistic model for the motion of the swimmer?

1.11 — Question 5. The income £I made by a company on various cleaning services they provide may be modelled by the equation $I = \frac{5}{2}x(40 - x)$ where £x is the price charged for each individual service.

a) Sketch the model for $0 \leq x \leq 40$.

b) Rearrange the model into the form $I = p - \frac{5}{2}(x - q)^2$ where p and q are constants to be found.

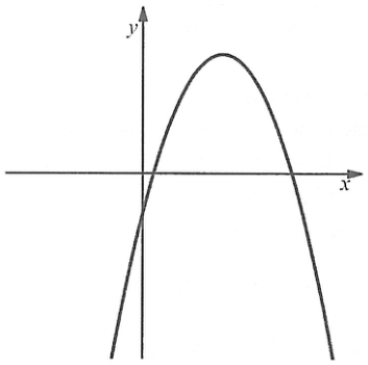
According to the model:

c) Find the maximum income that is made.

d) Find the price of the service that gives the maximum income.

e) Find the prices which give an income of £600.

1.11 — Question 6. The profit of a business selling books depends on the price at which each book is sold. The equation $y = \frac{15}{2}x - \frac{25}{2} - \frac{5}{8}x^2$ is used to model the relationship between the profit and the price where £x is the selling price of each book and y pence is the profit achieved at that price. A sketch of the curve is shown in the diagram below:



Find

a) the selling price that generates the maximum profit.

b) the value of the maximum profit.

c) the selling price that gives no profit.

d) the loss if the books are given away.

1.11 — Question 7. The growth of two flowers may be modelled by quadratic curves as follow:
Flower A: height y cm after x days is given by $y = -\frac{2}{45}x^2 + \frac{8}{3}x$.
Flower B: height y cm after x days is given by $y = -\frac{1}{10}x^2 + 3x + \frac{15}{2}$.

a) Calculate the number of days which have passed when they are the same height.
b) Find the maximum height of the curve representing flower B.
c) Explain why the growth of the flowers may not best be modelled by these models.

1.11 — Question 8. A ball is projected from the ground. It reaches a maximum height of 10 metres at 2 seconds since being projected. Find a model for the trajectory of the ball in the form $h = at^2 + bt$ where h is height (metres), t is time (seconds) and a and b are constants to be found.

1.11 — Question 9. A stone is projected from the ground. It starts at a horizontal position of a metres from a wall. It is modelled by $y = -\frac{1}{8}x^2 + \frac{11}{8}x - \frac{57}{32}$ where x and y are the horizontal and vertical distance in metres respectively from the bottom of the wall.

a) Find the value of a.
b) Find the total horizontal distance travelled by the ball before it hits the ground.
c) Find the maximum height that the ball reaches.
d) A tree of height 1.2 m is 7.5 m from the first wall, does the stone hit the tree?

1.11 — Question 10. An arch has a span of 350m and a maximum vertical height of 140m above ground level. Let x be the horizontal distance from the left-most point of the arch in metres and y be the vertical distance above the ground. Find a model in the form $y = \frac{p}{q}x(r - x)$ where p, q and r are integers to be found.

1.13 — Question 1. Simplify the following as far as possible:

a) $\frac{(x+4)(x+3)}{x+3}$
b) $\frac{x^2+11x+30}{x+5}$
c) $\frac{x^2+5x}{x^2+7x+10}$
d) $\frac{x^2+2x-35}{x^2+4x-21}$
e) $\frac{2x^2+x-1}{2x^2+15x-8}$
f) $\frac{3x^2+7x+4}{9x^2-16}$
g) $\frac{2x^2+9x+4}{4x^3-8x^2-5x}$
h) $\frac{2x^2+17x+30}{2x^3+11x^2+15x}$

1.13 — Question 2. Write as a single fully simplified fraction:

a) $\frac{1}{x}+\frac{3}{2x}$
b) $\frac{1}{x^2}+\frac{1}{x}+\frac{3}{5}$
c) $\frac{1}{x}+\frac{1}{x+1}$
d) $\frac{1}{x+2}+\frac{2}{x+4}$
e) $\frac{3}{x+1}-\frac{1}{x-3}$
f) $\frac{5}{x+2}+\frac{7}{2x-1}$
g) $\frac{3}{2(x+5)}+\frac{1}{4(x-3)}$
h) $\frac{x}{x^2-4}+\frac{5}{x+2}$

1.13 — Question 3. Simplify and fully factorise the following:

a) $\frac{x^2-5x+4}{x^2-x} \times \frac{2x+3}{5x-20}$

b) $\frac{x-3}{3x^2-6x} \times \frac{3x^2-12}{x^2-9}$

c) $\frac{16x^2-4}{3x-1} \times \frac{3x^2+14x-5}{4x+2}$

d) $\frac{x+2}{x+3} \div \frac{x^2+3x+2}{2x^2-2x-12}$

1.13 — Question 4. Fully simplify the following:

a) $\frac{\frac{1}{x}}{\frac{1}{x+1}}$

b) $\frac{\frac{1}{x-2}}{\frac{1}{2x}}$

c) $\frac{\frac{1}{x+3}+\frac{1}{x-5}}{\frac{x-5}{x+3}}$

d) $\frac{\frac{1}{x+4}+\frac{1}{x-1}}{\frac{x+2}{x-1}}$

e) $\frac{(3x+1)(5x-1)+4(5x-1)}{\frac{5x-1}{3x+1}}$

1.13 — Question 5. Prove each of the following:

a) $\frac{3}{x+2} + \frac{x-2}{(x+2)(x+3)} \equiv \frac{4x+7}{(x+2)(x+3)}$

b) $\frac{2x^2-x-1}{x^2-1} - \frac{1}{x^2+x} \equiv \frac{2x-1}{x}$

c) $1 - \frac{1}{x-3} + \frac{2}{x^2-9} \equiv \frac{x^2-x-10}{(x-3)(x+3)}$

d) $\frac{9-4x}{2(x-1)} + \frac{1}{x+3} + 2 \equiv \frac{7x+13}{2(x-1)(x+3)}$

1.14 — Question 1. Expand each of the following:

a) $(x-1)(2x+1)(7x+3)$

b) $(2x-1)(3x-4)(5x-2)$

c) $(4x+3)(2x+1)(3x-2)$

1.14 — Question 2. Fully factorise each of the following:

a) $8x^3 - 12x^2 - 28x$

b) $10x^4 - 20x^3 - 30x^2$

c) $6x^4 - 45x^3 - 24x^2$

1.14 — Question 3. Fully factorise each of the following:

a) $2x^2y^3z - 6xy^4z^2$

b) $4x^2y^3z^4 + 14x^3y^2z^5$

c) $3(x-1)^3y^2z - 9(x-1)^2yz^3$

d) $9x^2(2y-1)^3z - 15x^3(2y-1)z^2$

e) $8x^2(3y-2)^3(z+1) - 4x^3(3y-2)(z+1)^2$

1.14 — Question 4. State the degree of the following polynomials:

a) $6x^4 - 9x^2 + 2$

b) $4x^2 - 3x^5 + 1$

c) $8 + 4x^3 - x^6$

1.14 — Question 5. Expand each of the following:

a) $(x + 2)(x^2 - 3x + 1)$

b) $(2x - 1)(x^3 + 2x^2 - x + 3)$

c) $(x^2 + 2x + 3)(3x^2 - x + 2)$

d) $(2x^2 - 4x + 1)(x^2 + 3x - 2)$

1.14 — Question 6. Simplify the fractions:

a) $\frac{3x^3+5x^2}{x}$
b) $\frac{7x^3+9x^{10}}{3x}$
c) $\frac{(x+1)(x-5)}{x+1}$
d) $\frac{(x-2)^3}{x-2}$
e) $\frac{x+4}{(3x-1)(x+4)}$
f) $\frac{x-5}{2x^2-9x-5}$
g) $\frac{2x^2-15x+7}{2x^2+5x-3}$

1.14 — Question 7. Divide

a) $4x^3 - 22x^2 + 34x - 12$ by $2x - 4$
b) $x^4 + 6x^3 + 7x^2 + x + 6$ by $x + 2$
c) $6x^5 - 15x^4 + 9x^3 - 36x^2 + 72x - 24$ by $3x - 6$
d) $32x^4 - 12x^3 + x^2 + 20x - 5$ by $4x - 1$
e) $2x^6 + x^5 - 3x^4 + 4x + 6$ by $2x + 3$
f) $x^5 + 3x^4 + 4x^3 - 2x^2 - 4x - 5$ by $x^2 + x + 1$
g) $x^7 - 5x^6 + x^5 - 4x^4 - 8x^3 + 13x^2 - 8x + 12$ by $x^2 + 1$

1.14 — Question 8. Divide

a) $x^3 + 9x^2 + 19x + 6$ by $x + 1$

b) $4x^3 - 10x^2 - 12$ by $2x - 1$

c) $x^4 + 2x^3 - x^2 + x + 16$ by $x + 2$

d) $x^5 - 4x^3 + x^2 + 10$ by $x^2 - 1$

1.14 — Question 9. Given $(x - 5)$ is a factor of $f(x)$, where $f(x) = x^3 - 3x^2 - 8x - 10$,

a) Write $f(x)$ as a product of a quadratic and a linear expression.

b) Show that $f(x) = 0$ only has one real solution.

1.14 — Question 10. Given $(x+2)$ is a factor of $f(x)$, where $f(x) = 2x^3 - 7x^2 - 10x + 24$,

a) Write $f(x)$ as a product of a quadratic and a linear expression.
b) Fully factorise $f(x)$.
c) Find the solutions to $f(x) = 0$.
d) Write down the roots of $f(x-2) = 0$.

1.14 — Question 11. Given $(2x+1)$ is a factor of $f(x)$, where $f(x) = 2x^3 - x^2 + 35x + 18$,

a) Write $f(x)$ as a product of a quadratic and a linear root.
b) Show that $f(x) = 0$ only has one real solution.
c) State the one real solution to $f(x) = 0$.
d) Write down the real root of $f(x-4) = 0$.

1.14 — Question 12. Given $f(x)$ and $g(x)$ both have a common factor of $(x-1)$ where $f(x) = x^3 + 4x^2 - 5x$ and $g(x) = x^3 - 3x + 2$,

a) Fully factorise $f(x)$ and $g(x)$.

b) Show that $(x-1)$ is also a factor of $f(x) - g(x)$.

1.14 — Question 13. The Factor Theorem Use the factor theorem to prove that

a) $(x-3)$ is a factor of $x^3 - 10x^2 + 13x + 24$

b) $(x+5)$ is a factor of $x^3 - 5x^2 - 10x + 200$

c) $(4x-1)$ is a factor of $4x^3 - 21x^2 + 29x - 6$

1.14 — Question 14. In each case, use the factor theorem to prove the given factor is a factor and hence fully factorise the expression:

a) $(x-6)$ is a factor of $2x^3-11x^2-9x+18$

b) $(2x+5)$ is a factor of $6x^3+19x^2+8x-5$

1.14 — Question 15. It is given that $(x+1)$ is a factor of $(x+3)(x^2-kx+4)-20$ where k is a positive constant. Find the value of k.

1.14 — Question 16. It is given that $(x+3)$ is a factor of $f(x) = 2x^3 + 7x^2 - px - 18$ where p is a positive constant.

a) Find the value of p.

b) Fully factorise $f(x)$.

1.14 — Question 17. It is given that $(8x + 15)$ is a factor of $f(x) = 32x^3 + px^2 + 219x + 45$ where p is a positive constant.

a) Find the value of p.

b) Fully factorise $f(x)$.

1.14 — Question 18. It is given that $(x+1)$ is a factor of $f(x) = 2x^3 + 11x^2 - px - 2p$ where p is a positive constant.

a) Find the value of p.

b) Fully factorise $f(x)$.

1.14 — Question 19. Given that $(x+2)$ and $(x-5)$ are factors of $f(x) = px^3 - 5x^2 - 23x - q$ where p and q are positive constants.

a) Find the values of p and q.

b) Fully factorise $f(x)$.

1.14 — Question 20. Given that $(x+3)$, $(x-1)$ and $(2x-1)$ are factors of $f(x) = px^4 + qx^3 + rx^2 - 37x + 15$ where p, q and r are constants. Find the values of p, q and r.

1.14 — Question 21. It is known that $(x+9)$ is a factor of $f(x) = 3x^4 + 59x^3 + px^2 + 369x - 162$.

a) Find the value of p.

b) Fully factorise $f(x)$.

1.14 — Question 22. It is given that $x^2 + bx + c$ has a factor of $(x - 1)$ and $x^2 + px + q$ has a factor of $(x + 2)$. Show that $5 + b + c - 2p + q = 0$.

1.14 — Question 23. Show that $135k^3 + 270k^2 + 165k + 30$ is always a multiple of 30 when k is a positive, whole number.

1.14 — Question 24. Prerequisite knowledge for Q24-Q28: Graph Sketching $f(x) = x^3 + 2x^2 + 2x - 20$

a) Use the factor theorem to show that $(x - 2)$ is a factor of $f(x)$.

b) Prove that the graph of $f(x)$ intersects the x-axis only once and state its coordinates.

c) Given that the point of inflection is to the left of the root of the curve, sketch $y = f(x)$.

1.14 — Question 25. It is given that $(x+3)$ is a factor of $f(x) = x^3 - 7x^2 - px + 75$ where p is a positive constant.

a) Find the value of p.

b) Find the solutions to $f(x) = 0$.

c) Sketch the graph of $y = f(x)$.

1.14 — Question 26. Given that $(x-3)$ and $(x+1)$ are factors of $f(x) = x^3 - (2+k)x^2 + (2k-3)x + 3k$ where $0 < k < 3$. Sketch the graph of $f(x)$.

1.14 — Question 27. It is given that $f(x) = x^3 - (2k+4)x^2 + (k^2+8k)x - 4k^2$ where k is a positive constant.

a) Prove that $(x-4)$ is a factor of $f(x)$.

b) Given $0 < k < 4$, sketch the graph of $f(x)$.

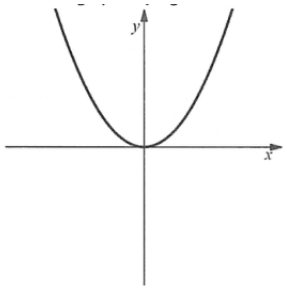
1.14 — Question 28.

a) Divide $x^3 - x^2 + 5$ by $x - 1$.

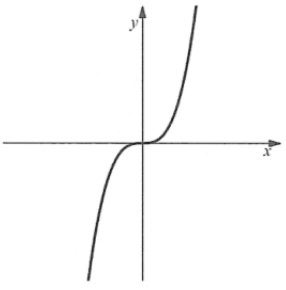
b) Hence, sketch the graph of $y = \frac{x^3 - x^2 + 5}{x - 1}$.

1.17 — Question 1. It is given that y is inversely proportional to x^3. State which of the following diagrams could represent the graph of y against x.

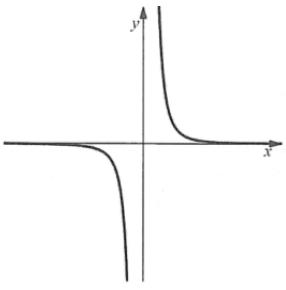
A



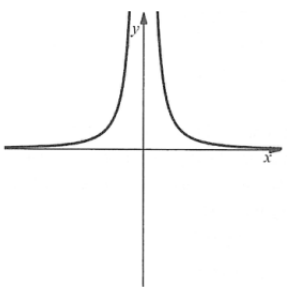
B



C



D



1.17 — Question 2. In each of these cases, write down an equation for y in terms of x and k, where k is a constant of proportionality:

a) y is directly proportional to the square of x
b) y is inversely proportional to x
c) y is directly proportional to the cube root of x
d) y is inversely proportional to the cube of $x + 1$

1.17 — Question 3. Given that y is directly proportional to x^2, and when $x = 5$, $y = 100$, find y when $x = 10$.

1.17 — Question 4. Given that y is inversely proportional to $\sqrt[4]{x}$, and when $x = 81$, $y = 18$, find x when $y = 180$.

1.17 — Question 5. The volume of gas, V litres, that a gas canister can hold is directly proportional to the temperature, T Kelvin. It is given that when $T = 300$, $V = 24$. Determine the volume of gas when the temperature is 400 Kelvin.

1.17 — Question 6. The time taken, t hours, to complete a job is inversely proportional to the number of workers, n, employed. It is given that when $n = 5$, $t = 8$. Determine the number of workers required to complete the job in 4 hours.

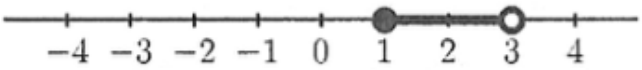
1.17 — Question 7. The rate of reaction, R mol/s, of a certain chemical reaction is directly proportional to the concentration C mol/L of a reactant. It is given that when $C = 0.5$, $R = 2$. Determine the concentration that will give a reaction rate of 6 mol/s.

1.17 — Question 8. The intensity of light, I lumens, from a light source is inversely proportional to the square of the distance d metres from the source. It is given that when $d = 2$, $I = 400$. Determine the distance that will give a light intensity of 100 lumens.

1.17 — Question 9. It is given that y is directly proportional to $\sqrt{x}$, and that z is inversely proportional to y^{10}. Given that $z = 2$ when $x = 4$, find x when $z = \frac{1}{16}$.

1.20 — Question 1. For each of the following find the values which have been identified, using set notation:

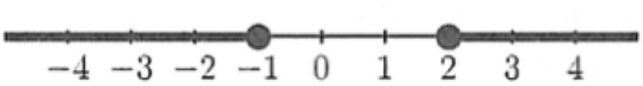
a)



b)



c)



d)



1.20 — Question 2. For each of the following find the values which have been identified, using interval notation:

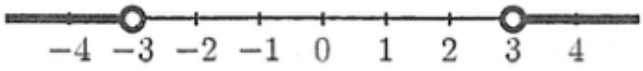
a)



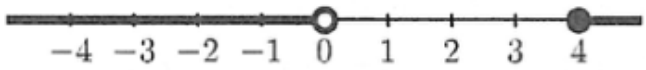
b)



c)



d)



1.20 — Question 3. Draw a number line to show thes values of real number x which belong in the following sets:

a) $\{x : x \leq -3\} \cup \{x : x > 2\}$
b) $\{x : x < 1\} \cup \{x : x \geq 2\}$
c) $\{x : x \geq -2\} \cap \{x : x < 3\}$
d) $\{x : x > -1\} \cap \{x : x \leq 2\}$
e) $\{x : x > 1\} \cap \{x : x < -3\}$

1.20 — Question 4. Solve the following inequalities giving your answer in set notation:

a) $3x - 8 < 4$
b) $6 - 2x \leq 15$
c) $6 + 5x > 2 - 3x$
d) $1 + 9x \leq 6 - x$
e) $-2 \leq 3x + 4 \leq 10$
f) $-5 < 5x - 8 < 22$
g) $-1 < 2x + 1 < 1$
h) $-1 < ax + b < 1$ where $a > 0$
i) $-1 < ax + b < 1$ where $a < 0$

1.20 — Question 5. Solve the following inequalities, writing your answer in interval notation:

a) $\frac{x+14}{3} < 2(x-1)$

b) $\frac{4(x+1)}{3} \geq \frac{x-6}{2}$

c) $\frac{4x+1}{3} - \frac{2x-1}{2} \leq 1$

1.20 — Question 6. Solve the following inequalities giving your answer in set notation:

a) $(x+2)(x-4) < 0$

b) $(x+1)(x+7) \geq 0$

c) $x^2 + 3x - 10 \leq 0$

d) $x^2 + 5x + 6 > 0$

e) $-(x+3)(x-8) < 0$

f) $(2-x)(4+x) \geq 0$

g) $x^2 + 3x - 1 > 0$

h) $x^2 - 2x + 1 \geq 0$

i) $-x^2 + x - 3 > 0$

1.20 — Question 7. Solve the following inequalities giving your answer in interval notation:

a) $x^2 - x > 20$
b) $2x^2 < 10 - 8x$
c) $4x^2 + 4x \geq 35$
d) $6x^2 \leq 7x + 5$

1.20 — Question 8. Freddie tries to solve the inequality $2x - x^2 + 8 \leq 0$. His workings and answer are as follows:

$$\begin{aligned} x^2 - 2x - 8 &\leq 0 \\ (x+4)(x-2) &\leq 0 \\ x &= -2, x = 4 \end{aligned}$$

$$\{x : x < -2\} \cup \{x : x > 4\}$$

a) Identify all his mistakes.
b) Fully correct his workings.

1.20 — Question 9. Find the set of values of x for which the line $y = 2x + 3$ lies below the curve $y = x^2$ giving your answer in set notation.

1.20 — Question 10. Find the set of values of x for which the line $y = 19 - 10x$ lies above the curve $y = (2x + 1)^2$ giving your answer in interval notation.

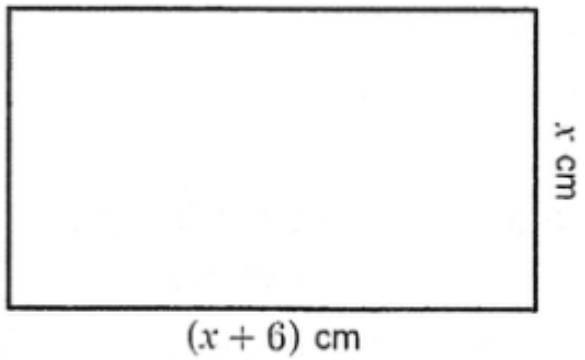
1.20 — Question 11. Solve $\frac{1}{x^2} < 529$ giving your answer in interval notation.

1.20 — Question 12. Solve $\frac{1}{x^2} \geq a^2$ where $a \neq 0$. Give your answer in terms of a and in set notation.

1.20 — Question 13. Re-write the following phrases into mathematical inequalities:

a) x is more than 12
b) x is less than 100
c) x is at most 5
d) x is at least 10
e) x is at least 50% more than y
f) x is at most 20% more than y

1.20 — Question 14. A picture frame of height x cm is 6 cm wider than it is tall. The width needs to be at least 75% more than its height.



a) Show that $x \leq 8$.

The back of the picture frame has an area of at most 160 cm^2.

b) Find the possible values of x giving your answer in set notation.
c) Hence state its maximum area.

1.20 — Question 15. Joe wants to install a rectangular swimming pool in his garden. He would like it to be 2 metres longer than it is wide but at least 8 metres long. He wants a pool that is less than 80 m^2 in area. Given the width is x metres, find the possible values of x giving your answer in set notation.

1.20 — Question 16. Prove that $x^2 + 3x > -4$ for all real values of x.

1.20 — Question 17. The equation $x^2 + kx + 9 = 0$ where k is a constant has real roots.

a) Show that $k^2 - 36 \geq 0$.

b) Hence find the set of possible values of k giving your answer in set notation.

1.20 — Question 18. Find the values of k for which each of the following equations have no real roots, giving your answer in set notation:

a) $x^2 + 3kx + 1 = 0$

b) $x^2 - kx + 5 - 2k = 0$

c) $(2k-1)x^2 + (2k+3)x + k = 0$, $k \neq \frac{1}{2}$

1.20 — Question 19. Find the set of values of k such that the curve $y = x^2 - 3kx + 8k$ lies entirely above the x-axis. Write your answer in set notation.

1.20 — Question 20. Find the set of values of k such that the curve $y = (k+2)x^2 + kx - 2k$ lies entirely above the x-axis. Write your answer in interval notation.

1.20 — Question 21. Find the set of values of k such that the curve $y = (k+1)x^2 - kx + k$ lies entirely below the x-axis. Write your answer in set notation.

1.20 — Question 22. Prerequisite knowledge for Q22: Circles Find the values of k such that the line $y = x + k$ intersects the circle with equation $x^2 + y^2 = 16$. Write your answer in set notation.

1.20 — Question 23. Prerequisite knowledge for Q23: Cubic Curves By first sketching the curve, solve the following inequalities, writing your answer in set notation:

a) $(x-3)(x+2)(x+1) > 0$
b) $(1-x)(x-5)(1+x) > 0$
c) $(x-6)(x-1)(2-x) \leq 0$
d) $x(5-x)(3+x) \geq 0$

1.20 — Question 24.

a) Sketch the curve $f(x) = x^2(x-2)$ labelling the points where the curve and the axes meet.
b) Hence find the set of values of x such that $f(x) \leq 0$ giving your answer in set notation.

1.20 — Question 25.

a) Sketch the curve $f(x) = (x+1)(x-3)^2$ labelling the points where the curve and the axes meet.

b) Hence find the set of values of x such that $f(x) > 0$ giving your answer in set notation.

1.20 — Question 26.

a) Sketch the curve $f(x) = x^2(x-a)$ where $a > 0$, labelling the points where the curve and the axes meet.

b) Hence find the set of values of x such that $f(x) > 0$ giving your answer in set notation.

1.20 — Question 27.

a) Sketch the curve $f(x) = (x+2)(x-a)^2$ where $a > 0$, labelling the points where the curve and the axes meet.

b) Hence find the set of values of x such that $f(x) \leq 0$ giving your answer in set notation.

1.20 — Question 28.

a) Sketching the graph of $y = (a-x)(x-b)^2$ where $a > 0$ and $0 < b < a$, labelling the points where the curve and the axes meet.

b) Find the values of x such that $(a-x)(x-b)^2 \geq 0$ giving your answer in set notation.

c) Find the values of x such that $(a-x)(x-b)^2 \leq 0$ giving your answer in interval notation.

1.20 — Question 29. Prerequisite knowledge for Q29: Factor Theorem Given $f(x) = 4x^3 - 24x^2 + 21x - 5$,

a) Use the factor theorem to show that $(x - 5)$ is a factor of $f(x)$.
b) Hence fully factorise $f(x)$.
c) Sketch $y = f(x)$.
d) Find the values for which $f(x) \leq 0$ giving your answer in interval notation.

1.20 — Question 30. Prerequisite knowledge for Q30-Q31: Graph Transformations Given $f(x) = x^3 + 4x^2 - 3x - 18$,

a) Show that $(x - 2)$ is a factor of $f(x)$.
b) Hence fully factorise $f(x)$.
c) Sketch $y = f(x)$.
d) Solve $f(2x) = 0$.
e) Hence, find the values for which $f(2x) > 0$ giving your answer in set notation.

1.20 — Question 31. A curve is given by $f(x) = -2x^3 + 18x^2 - 28x - 48$.

a) Fully factorise $f(x)$.

b) Hence, find the values for which $f(\frac{1}{2}x) > 0$ giving your answer in set notation.

1.20 — Question 32. Prerequisite knowledge for Q32-Q34: Exponentials and Logarithms Solve the following inequalities, writing your answer in set notation:

a) $e^x - 10 \leq 0$

b) $4^x - 5 \geq 0$

c) $(\frac{1}{3})^x > 9$

d) $(\frac{1}{5})^x < 1$

e) $0.4^x > 6.25$

1.20 — Question 33. Giving your answer in set notation, find the set of values of x for which $2\times 3^{2x}-3^{x}-28<0$.

1.20 — Question 34. Giving your answer in set notation, find the set of values of x for which $e^{x}-\frac{4}{e^{x}}>-3$.

1.21 — Question 1. Sketch and shade the region satisfied by

a) $y > 3x + 1$

b) $y \leq 2x - 5$

1.21 — Question 2. Sketch and shade the region satisfied by both

$$y \leq 2x - 1 \text{ and } x + 2y > 8$$

1.21 — Question 3. Sketch and shade the region satisfied by both

$$y > 3x + 8 \text{ and } x + 3y < 4$$

1.21 — Question 4. In each case, sketch a diagram and shade the region which satisfies the inequality:

a) $y \leq (x - 5)(x + 3)$
b) $y \geq (2x + 1)(x - 4)$
c) $y < x^2 + x - 2$
d) $y > 2x^2 - 7x - 15$

1.21 — Question 5. In each case, sketch a diagram and shade the region which satisfies the inequalities:

a) $y + x < -5$ and $y > x^2 + x - 20$

b) $y - 2x < 6$ and $y \leq -x^2 - x + 6$

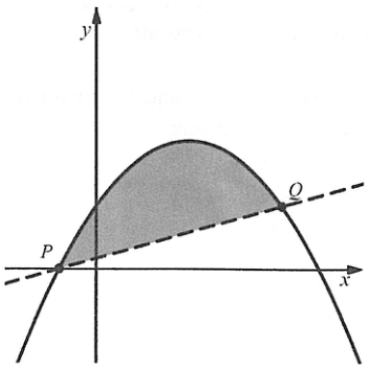
1.21 — Question 6. Sketch and shade the region satisfied by

$$3y \geq x, y + x < 4 \text{ and } y \geq (x - 2)^2$$

1.21 — Question 7. Sketch and shade the region satisfied by

$$y \geq x^2, x + y \leq 6 \text{ and } x \leq 0$$

1.21 — Question 8. The diagram shows the line $y = x + 1$ and the curve $y = 6 + 5x - x^2$



a) Write down the inequalities that define the shaded region.
b) Find the coordinates of the points P and Q
c) Solve the inequality

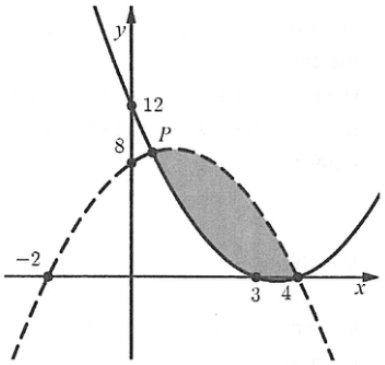
$$6 + 5x - x^2 > x + 1$$

giving your answer in set notation.

1.21 — Question 9. The diagram shows two curves of the form

$$y = ax^2 + bx + c$$

where a, b, c are non-zero constants.



a) Write down the inequalities that define the shaded region.
b) Find the coordinates of the points P

1.21 — Question 10.

a) Sketch and shade the region which satisfies both

$$y + 3 \geq x^2 \text{ and } y + 2x^2 < x + 1$$

b) Hence, solve the inequality

$$3x^2 - x - 4 < 0$$

giving your answer in set notation.

1.21 — Question 11. Prerequisite knowledge for Q11-Q12: Cubic Curves Sketch and shade the region which satisfies

$$y < x(x-2)(x+4)$$

1.21 — Question 12. Sketch and shade the region which satisfies

$$y \geq x^3 - 6x^2 + 11x - 6 \text{ and } y < 2x - 2$$

1.21 — Question 13. Prerequisite knowledge for Q13-Q14: Circles

a) Show that $x^2 + 2x + y^2 = 3$ is an equation of a circle, stating the centre and the radius.

b) Hence sketch a diagram and shade the region which satisfies

$$x^2 + 2x + y^2 < 3$$

1.21 — Question 14. Sketch a diagram and shade the region which satisfies both

$$(x-3)^2 + (y-1)^2 \leq 25 \text{ and } y > 3 - x$$

1.29 — Question 1. Without using your calculator, find the value of

a) $\frac{8!}{6!}$
b) $\frac{50!}{47!3!}$
c) $\frac{50!}{47!4!}$
d) $\binom{7}{6}$
e) $\binom{100}{98}$
f) $\binom{100}{99}$
g) ${}^{50}C_{49}$
h) ${}^{102}C_{102}$

1.29 — Question 2. Simplify

a) $\frac{n!}{(n-1)!}$
b) $\frac{n!}{2!(n-2)!}$
c) $\binom{n}{1}$
d) $\binom{n+1}{n}$
e) ${}^{n}C_{n}$
f) $\binom{2n-1}{2}$
g) $\binom{2n-4}{2n-6}$

1.29 — Question 3. Show that $10\binom{n}{5} = 21\binom{n}{3}$ simplifies to $n^2 - 7n - 30 = 0$ and hence find the value of n.

1.29 — Question 4. Solve the following equations:

a) $3 \times {}^nC_6 = 11 \times {}^nC_4$

b) $28 \times {}^nC_4 = 15 \times {}^nC_6$

c) $33 \times {}^nC_2 = 2 \times {}^nC_5$

1.29 — Question 5. Fully expand in ascending powers of x:

a) $(1+x)^6$
b) $(2+x)^5$
c) $(x+3)^6$
d) $(x-1)^4$

1.29 — Question 6. Fully expand in descending powers of x:

a) $(x-3)^5$
b) $(5-x)^6$
c) $(2x+1)^4$
d) $(3x-3)^5$
e) $(1+\frac{1}{2}x)^4$
f) $(2-\frac{x}{4})^5$
g) $(2x-\frac{1}{x})^4$

1.29 — Question 7. In ascending order, fully expand and simplify, $-5(3-2x)^4$.

1.29 — Question 8. In ascending order, fully expand and simplify, $(1+2x)(2-x)^4$.

1.29 — Question 9. In ascending order, fully expand and simplify, leaving your answers in exact form:

a) $(1+\sqrt{3})^4$
b) $(\sqrt{2}+\sqrt{5})^4$
c) $(\sqrt{7}-\sqrt{2})^5$

1.29 — Question 10. Find the x^2 coefficient in the expansion of:

a) $(x+5)^4$
b) $(2x-1)^5$
c) $(8+x)^5$
d) $(1-3x)^6$
e) $(1+2x)^{78}$

1.29 — Question 11. Find the x^2 coefficient in the expansion of:

a) $(x - \frac{1}{x})^4$

b) $(3x - \frac{2}{x})^6$

c) $(x - \frac{1}{x})^{18}$

1.29 — Question 12. By using the definition of $\binom{n}{r}$, show that $\binom{n}{5} = \binom{n-1}{4} + \binom{n-1}{5}$.

1.29 — Question 13. By using the definition of $\binom{n}{r}$, show that $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$.

1.29 — Question 14. Explain why in the expansion of $(1-3x)^9$, the terms with odd powers of x have negative coefficients.

1.29 — Question 15. Find the value of n in the expansion of $(1+x)^n$ where $n \geq 5$, given that the coefficient of x^2 and the coefficient of x^5 is the same.

1.29 — Question 16. Find the value of n in the expansion of $(1+x)^n$ where $n \geq 4$, given that the coefficient of x^4 is 13 times the coefficient of x^2.

1.29 — Question 17. Find the value of n in the expansion of $(1+2x)^n$ where $n \geq 2$, given that the coefficient of x^2 is 10 times the coefficient of x.

1.29 — Question 18. Find the value of n in the expansion of $(1+\frac{x}{2})^n$ where $n \geq 5$, given that the coefficient of x^4 is 2 times the coefficient of x^5.

1.29 — Question 19. Find the x^2 coefficient in the expansion of:

a) $x^2(x+\frac{1}{x})^{12}$

b) $(1-x)(1+4x)^{10}$

c) $(1-2x)(3-x)^9$

1.29 — Question 20.

a) Fully expand $(3+x)^5$

b) Hence simplify $(3-\sqrt{2})^5$ giving your answer in the form $a+b\sqrt{2}$ where a and b are constants.

1.29 — Question 21.

a) Fully expand $(2+5x)^4$

b) State the full expansion of $(2-5x)^4$

c) Hence fully simplify $(2+\sqrt{75})^4+(2-\sqrt{75})^4$

1.29 — Question 22.

a) Find the first 4 terms in ascending order of $(1+ax)^8$ where a is a positive constant.

b) Given that the coefficient is x^3 is 3 times the coefficient of x^2, find the value of a

1.29 — Question 23. Find the possible values of a in the expansion of $(a+2x)^6$ given the x^2 term has coefficient 4860.

1.29 — Question 24. Find the values of a and n such that $(1+ax)^n = 1+40x+700x^2+\cdots$ where $a, n \in \mathbb{Z}$.

1.29 — Question 25. Find the value of a given in the expansion of $(2x - \frac{1}{ax})^6$ the coefficient of x^4 is -12.

1.29 — Question 26.

a) In ascending order, find the expansion of $(2 - 7x)^5$

b) Given that in the expansion of $(1 + ax)(2 - 7x)^5$ the x term has coefficient of -464, find the value of a

1.29 — Question 27.

a) Find the first 3 terms in ascending order of $(1-2x)^9$

b) Hence find the value of 0.98^9 giving your answer to 3 decimal places.

1.29 — Question 28. Explain how you can use the expansion of $(3+\frac{2x}{5})^5$ to find the value of 2.96^5.

1.29 — Question 29.

a) Find the first 3 terms in ascending order of the expansions of $(1 - \frac{x}{2})^5$ and $(2 - \frac{3x}{4})^6$

b) Hence, estimate $0.995^5 + 1.9925^6$ giving your answer to 3 decimal places.

1.29 — Question 30. Without fully expanding $(x^2 + \frac{1}{x^4})^{12}$ explain why all the indices of the powers of x are multiples of 6.

1.29 — Question 31.

a) Fully expand $(1+3x)^4$

b) Hence state the full expansion of $(1-3x)^4$

c) State the minimum value of $f(x) = (1+3x)^4 + (1-3x)^4$

1.29 — Question 32.

a) Find the first 3 terms in ascending order in the expansion of $(5-2x)^6$

b) Hence determine the coefficient of the quadratic term in $(5-2x-6x^2)^6$

c) Use your answer to part a) to find the coefficient of the x^2 term of $(5+3x-2x^2)^6$

1.44 — Question 1. Solve the following equations, giving your answer in exact form:

a) $5x^4 = 3 - 2x^2$

b) $3x - 16\sqrt{x} = 35$

c) $2x^{\frac{2}{3}} + 7x^{\frac{1}{3}} = 15$

d) $\frac{6}{x^2} + \frac{7}{x} = 5$

e) $7^x + 7^{2x} = 2$

f) $4^x - 3 \times 2^{x-1} + \frac{1}{2} = 0$

g) $3^{3x+1} + 20 \times 3^{2x} - 7 \times 3^x = 0$

h) $2^{x+1} + 11 = \frac{6}{2^x}$

1.44 — Question 2. By first factorising, express $5^8 - 3^8$ as a product of its prime factors.

1.44 — Question 3. Find the solutions to

$$\frac{2x^4 + 2}{5} = x^2$$

1.44 — Question 4. Given

$$\mathrm{f}(x) = 8x^3 + \frac{1}{x^3}$$

solve $\mathrm{f}(x) = -9$

1.44 — Question 5. Given

$$\mathrm{f}(x) = 2x^6 + \frac{1}{x^3} \text{ and } \mathrm{g}(x) = x^3 + 2$$

solve $\mathrm{f}(x) = \mathrm{g}(x)$

1.44 — Question 6. Find the solutions to

$$\left(x - \frac{4}{x}\right)^2 + 8 = 6\left(x - \frac{4}{x}\right)$$

1.44 — Question 7. Find the solutions to

$$\frac{5x^2}{\sqrt{x}} + 22 + \frac{24}{\sqrt{x}} = 12x + \frac{39x}{\sqrt{x}}$$

1.44 — Question 8. Find the solutions to

$$3^{2x+1} + 20 \times 3^x + \frac{75}{3^{2x}} = 22 + \frac{220}{3^x}$$

1.44 — Question 9.

Prerequisite knowledge for Q9–Q12: Exponentials and Logarithms

Solve the following equations, giving your answer in exact form:

a) $3 \times 2^{2x} + 7 \times 2^x = 6$

b) $2e^{3x} + 3e^{2x} - 17e^x + 12 = 0$

c) $4 \times 3^x + 13 = \frac{2}{3^x} + \frac{15}{3^{2x}}$

1.44 — Question 10.

a) Solve

$$\frac{4}{(x-2)^2} - \frac{7}{x-2} = 2$$

b) Hence, solve

$$\frac{4}{(5^x-2)^2} - \frac{7}{5^x-2} = 2$$

giving your answer in exact form.

1.44 — Question 11. Solve the following equations, giving your answer in exact form:

a) $\log_2(x+3) = 2\log_2 x - \log_2 4$

b) $\log_4(5x+2) = \log_4(13x-6) - 2\log_4 x$

c)

$$-1 + \log_5(x^4+4)$$
$$= \log_5 x + \log_5(3x+14-5x^2) - \log_5 6$$

1.44 — Question 12. Solve for the following equation giving your answer in exact form

$$\ln x + \frac{4}{\ln x} = \frac{\ln(x^4)}{\ln x}$$

1.44 — Question 13.

Prerequisite knowledge for Q13–Q14: Solving Trigonometric Equations

Solve

$$\sin^2 x = \frac{7 - 13\sin x}{2}$$

for $0 \leq x \leq 360°$

1.44 — Question 14. Solve

$$\frac{\cos^2 x(3\cos x + 11)}{20} = \cos x$$

for $0 \leq x \leq 360°$